

# DMRG class at BME – projects ("A" exam) , literature ("B" exam), and exam topics ("C" exam)

Miklós Werner, Kornél Kapás & Örs Legeza

Spring semester 2026

## Projects (TYPE "A" exam)

You may choose **one** of the projects below.

### 1. Symmetry Protected Topological order in the anisotropic Heisenberg Chain

Consider the  $S = 1$  Heisenberg chain with uniaxial anisotropy,

$$H = \sum_i \left( J \vec{S}_i \cdot \vec{S}_{i+1} + D(S_i^z)^2 \right) ,$$

where  $S_i^{x,y,z}$  are the  $3 \times 3$  spin matrices of an  $S = 1$  spin.

- Construct the MPO form of the Hamiltonian.
- Calculate the ground state for various anisotropy strengths  $D/J \in [0, 2.5]$ , using the infinite-chain (chain growing) DMRG algorithm. Approximate the energy density ( $\epsilon = E/L$ ) in the  $L \rightarrow \infty$  limit? (*Hint: instead of dividing the energy by the actual length, it is better to calculate the 'bond-energy'  $\epsilon = \langle J \vec{S}_i \cdot \vec{S}_{i+1} + D/2(S_i^z)^2 + D/2(S_{i+1}^z)^2 \rangle$* )
- From the Schmidt values, calculate the von Neumann entropy between the two half systems, and plot the  $S(D/J)$  function.
- Consider the Schmidt-spectrum of the ground state. What do you observe for small  $D/J$ ? And for large  $D/J$ ? Where is the transition point?
- Investigate, how your results depend on the MPS bond dimension  $M$ .
- (EXTRA TASK!) You may also calculate the energy gap by requiring eigs() to produce 2 eigenstates. (*Hint: if calculating 2 eigenstates simultaneously in DMRG, it is better to diagonalize the mixture of the block reduced density matrices, i.e.  $\rho_{\text{left}} = 0.5\rho_{\text{left}}^{\text{GS}} + 0.5\rho_{\text{left}}^{\text{exc}}$  to form better block states both in the left and the right.*)

### 2. Ground state of the $S = 1$ bilinear-biquadratic model

Consider the  $S = 1$  "bilinear-biquadratic" model,

$$H = \sum_i \left( \cos(\theta) \vec{S}_i \cdot \vec{S}_{i+1} + \sin(\theta) (\vec{S}_i \cdot \vec{S}_{i+1})^2 \right) ,$$

where  $\vec{S}_i = (S_i^x, S_i^y, S_i^z)$  and  $S_i^{x,y,z}$  are the  $3 \times 3$  spin matrices of an  $S = 1$  spin.

- Construct the MPO form of the Hamiltonian. What is the bond-dimension of the MPO?
- Calculate the ground state as a function of  $\theta \in [-\pi/2, \pi/2]$  using the infinite chain (chain-growing) DMRG algorithm. Approximate the energy density ( $\epsilon = E/L$ ) in the  $L \rightarrow \infty$  limit? (*Hint: instead of dividing the energy by the actual length, it is better to calculate the 'bond-energy'  $\epsilon = \langle \cos(\theta) \vec{S}_i \cdot \vec{S}_{i+1} + \sin(\theta) (\vec{S}_i \cdot \vec{S}_{i+1})^2 \rangle$* )

- From the Schmidt values, calculate the von Neumann entropy between the two half systems, and plot the  $S(\theta)$  function.
- Consider the following points:  $\theta = \{0, \arctan(1/3), \pm\pi/4, -\pi/2\}$  How do the entanglement entropy depend on the system size  $L$  in these points? Calculate also the  $\langle \vec{S}_i \cdot \vec{S}_{i+1} \rangle$  nearest neighbor correlator at the two central sites as a function of  $L$ . What did you learn about the ground state?
- Investigate, how your results depend on the MPS bond dimension  $M$ .

### 3. Information scrambling, out-of-time-ordered correlators (OTOC's)

Consider the mixed-field Ising chain,

$$H = - \sum_i (J Z_i Z_{i+1} + g Z_i + h X_i)$$

with  $J = 1$ ,  $h = -1.05$ , and  $g = 0.5$ , where  $Z_i$  and  $X_i$  are the usual spin-1/2 spin operators (Pauli matrices). Consider a chain of length  $L \approx 20 - 40$ . The goal of this project is to calculate the so-called out-of-time-ordered correlator (OTOC) of the  $Z$  operators,

$$C(r, t) = \langle \Psi | [Z_{L/2}(t), Z_{L/2+r}(0)]^\dagger [Z_{L/2}(t), Z_{L/2+r}(0)] | \Psi \rangle ,$$

where  $Z_i(t) = e^{iHt} Z_i e^{-iHt}$  is the time-dependent operator in the Heisenberg picture,  $[A, B] = AB - BA$  is the commutator, and  $|\Psi\rangle$  is a random product state in the  $Z$ -basis, i.e.,  $|\Psi\rangle \sim |\uparrow\uparrow\downarrow\uparrow\downarrow \dots \downarrow\downarrow\rangle$  for some random spin configuration  $\{\sigma_i\}$ .

Note that  $C(r, t) = (\langle \Psi_{12}^r(t) | - \langle \Psi_{21}^r(t) |) (| \Psi_{12}^r(t) \rangle - | \Psi_{21}^r(t) \rangle)$ , with

$$| \Psi_{12}^r(t) \rangle = e^{iHt} Z_{L/2} e^{-iHt} Z_{L/2+r} | \Psi \rangle ,$$

$$| \Psi_{21}^r(t) \rangle = Z_{L/2+r} e^{iHt} Z_{L/2} e^{-iHt} | \Psi \rangle .$$

In the "12" case, the action by  $Z_{L/2+r}$  is trivial as the initial state is an eigenstate of  $Z_{L/2+r}$ , i.e., we only get a  $\pm 1$  prefactor depending on  $r$  and the initial spin configuration. Therefore, in both cases, first we need to evolve the state  $|\Psi\rangle$  to  $t$ , then act by  $Z_{L/2}$ , then evolve back in time (with  $-dt$  steps) to 0. In case of the "12" branch, we then only need to apply the  $\pm 1$  factor depending on  $r$  and the initial configuration, while in case of "21" we need to apply  $Z_{L/2+r}$  on the state.

- Construct TEBD on the finite chain of length  $L$  with open boundary conditions.
- Use time-steps  $dt \approx 0.05 - 0.1$  and bond dimensions  $M \lesssim 100 - 200$  depending on your machine. (run the calculations at 2-3 different  $M$  values)
- Determine  $C(r, t)$  up to times  $T_{\max} \approx 10$ , and plot it on a heatmap.

### 4. Dynamical Phase Transitions, Loschmidt-echo

Consider the transverse field Ising model,

$$H = - \sum_i (J Z_i Z_{i+1} + h X_i) .$$

In this model, a quantum phase transition (QPT) occurs at  $h/J = 1$ . We initialize the system in the ground state at  $H(h_0)$ . At  $t = 0$  we change the transverse field to  $h_1$  (sudden global quench), leading to a time-dependent quantum state  $|\Psi(t)\rangle$ . The Loschmidt-Echo rate function in the thermodynamic limit is defined as

$$\lambda(t) = - \lim_{L \rightarrow \infty} \frac{1}{L} \ln |\langle \Psi(0) | \Psi(t) \rangle|^2 .$$

Our goal is to extract this function by simulating the dynamics using the infinite-chain TEBD (iTEBD) algorithm.

The initial field is set to  $h_0 = 0$ ; therefore, the initial state is a product (Neél state:  $\dots \uparrow\downarrow\uparrow\downarrow \dots$ ).

Note that in iTEBD, you have only two matrices (A & B) in the MPS, and  $\lambda(t)$  is related to the largest magnitude eigenvalue of the transfer matrix

$$T_{\alpha\tilde{\alpha}\beta\tilde{\beta}} = \sum_{\sigma_1, \sigma_2} \sum_{\gamma, \tilde{\gamma}} A(t)_{\alpha\gamma}^{\sigma_1} B(t)_{\gamma\beta}^{\sigma_2} \left( A(0)_{\tilde{\alpha}\tilde{\gamma}}^{\sigma_1} \right)^* \left( B(0)_{\tilde{\gamma}\tilde{\beta}}^{\sigma_2} \right)^* .$$

As the initial state is a product,  $\dim(\tilde{\alpha}) = \dim(\tilde{\beta}) = \dim(\tilde{\gamma}) = 1$ , i.e.,  $T$  is an  $M \times M$  matrix, whose largest magnitude eigenvalue at time  $t$  is  $\tau(t)$ . The overlap of a long segment of length  $L$  is then  $\tau^{L/2}$ , therefore,

$$\lambda(t) = - \lim_{L \rightarrow \infty} \frac{1}{L} \ln |\tau(t)|^L = - \ln |\tau(t)|$$

- Set up an iTEBD calculation with the Neél state as an initial state.
- Evolve the state for an external field  $h_1$ , and at every timestep  $t$ , form the  $T$  transfer matrix.
- By the diagonalization of the  $T$  transfer matrix, determine the Loschmidt echo rate function  $\lambda(t)$ .
- Plot and analyze  $\lambda(t)$ . How does it look like, if  $h_1 < 1$ ? And if  $h_1 > 1$ ?
- Perform your calculations for bond dimensions  $M \lesssim 200$  depending on your computer. How do your results depend on  $M$ ?
- Plot also the half-chain entropy as a function of  $t$  for different  $M$ 's. Up to what times are your results reliable?

## Literature tasks (TYPE "B" exam)

If you decide to get a grade by reading literature, select **one** of the following advanced topics, read the highlighted papers. On the exam, you are supposed to present the content. (You may take notes/perform calculations, that you can use at the exam).

### 1. Projected Entangled Pair States

- F. Verstraete and J. I. Cirac, "Renormalization algorithms for Quantum-Many Body Systems in two and higher dimensions", arXiv:0407066 (2004).
- R. Orús, "A practical introduction to tensor networks: Matrix product states and projected entangled pair states", Annals of Physics 349, 117–158 (2014).

### 2. Multi-scale Entanglement Renormalization Ansatz

- G. Vidal, "Class of Quantum Many-Body States That Can Be Efficiently Simulated", Phys. Rev. Lett. 101, 110501 (2008).
- G. Evenbly and G. Vidal, "Algorithms for entanglement renormalization", Phys. Rev. B 79, 144108 (2009).

### 3. Non-Abelian symmetries in TEBD

- M. A. Werner, C. P. Moca, Ö. Legeza, M. Kormos, and G. Zaránd, "Spin fluctuations after quantum quenches in the  $S = 1$  Haldane chain: Numerical validation of the semi-semiclassical theory", Phys. Rev. B 100, 035401 (2019).

- M. A. Werner, C. P. Moca, Ö. Legeza, and G. Zaránd, "Quantum quench and charge oscillations in the SU(3) Hubbard model: A test of time evolving block decimation with general non-Abelian symmetries", Phys. Rev. B 102, 155108 (2020).

#### 4. Controlled bond expansion

- Andreas Gleis, Jheng-Wei Li, and Jan von Delft, "Controlled Bond Expansion for Density Matrix Renormalization Group Ground State Search at Single-Site Costs", Phys. Rev. Lett. 130, 246402 (2023).
- Jheng-Wei Li, Andreas Gleis, and Jan von Delft, "Time-Dependent Variational Principle with Controlled Bond Expansion for Matrix Product States", Phys. Rev. Lett. 133, 026401 (2024)

### Exam topics (TYPE "C" (old school) exam)

1. Exact Diagonalization Techniques. Construction of lattice Hamiltonians (Kronecker products). Fermionic models and the Jordan-Wigner transformation. Full diagonalization / finding selected eigenvectors. The Lanczos algorithm. Symmetries and Quantum numbers.
2. RG based diagonalization. Selection of 'important' states of subsystems, and truncation. The Block-RG (BRG) algorithm for the transverse field Ising model. Convergence of BRG in block size, and limitations of the algorithm. The Numerical Renormalization Group (NRG) algorithm. Density matrix-based truncation, and the original (infinite-chain) DMRG algorithm.
3. Schmidt-decomposition, and truncation based on Schmidt values. Entanglement entropy and bond dimension. Equivalence with Density matrix-based truncation. Matrix Product States (MPS). Gauge freedom and canonical forms. Left canonical, right canonical, and mixed canonical forms. Abelian quantum numbers and block structure of MPS matrices.
4. The Density Matrix Renormalization Group (DMRG) algorithm. Infinite chain algorithm and finite chain optimization (sweeping). The 2-site iteration step, the effective Hamiltonian. Matrix Product Operator formalism, the XXZ spin chain and its MPO Hamiltonian. DMRG as a variational algorithm: 1-site and 2-site schemes. MPO for fermionic models (Jordan-Wigner  $\phi$ -factors in the MPO).
5. Real-time evolution of MPS states. One-dimensional nearest neighbor chains and the time-evolving block decimation (TEBD) algorithm. Entanglement barrier and maximal simulation time. TEBD for longer-ranged interactions – limitations of TEBD. Algorithms for long-ranged interactions: the basis update and Galerkin (BUG) method, and the 2-site time dependent variation principle (TDVP) algorithm.
6. Advanced topics. (i) Open quantum systems, MPS-based solution of the Lindblad equation. Vectorization of the density matrix. The triviality of the  $T = \infty$  state. (ii) Ordering optimization of MPS based in 2-site mutual info. The idea of fermionic mode optimization. Local mode optimization and its limitations. Global mode optimization with mutual info based reorderings, and with systematic use of swap gates. Mode optimization beyond the active space: the self-consistent field (SCF) approach.